

Appendix

Overview of Appendix

In the appendix, we provide details of propagation generation in Section A, the experimental setup in Section B, and present supplementary experimental results in Section C.

A Details of LLM-based Synthetic Propagation Generation

We describe the algorithm to generate a synthetic propagation graph under the specific attribute with large language models (LLMs). For English news, the assignments of Age, Gender, Education, and Politics are adapted from Wan *et al.* [2024]; Registration from Shu *et al.* [2019b]; Occupation from Motaleb *et al.* [2021]; and Location from the geographic regions of the United States¹. For the news in Chinese, the assignments for some attributes are different from those for English news, as for the assignment of the Location, we refer to the geographical divisions of China². The assignment of Politics is adapted from the three-dimensional political spectrum³. The specific assignments of user attributes are summarized in Table 1. Formally, given the generation size m and the targeted news n , we can obtain the synthetic propagation graph $G = (V, E)$ by Algorithm 1.

For the generation of the propagation under some user attributes, we uniformly sample each value within the attribute to form a user profile ($\psi()$). Then we randomly choose to comment on the news ($\phi_1()$) or reply to the existing comments ($\phi_2()$). During the generation, we record the user profile and the generated comment (the current node), as well as the comment or news replied to by the comment (parent node), to form a synthetic propagation graph for a piece of news. Specifically, $\phi_1()$ and $\phi_2()$ output prompts for LLMs to role-play are as follows:

¹https://en.wikipedia.org/wiki/List_of_regions_of_the_United_States

²https://en.wikipedia.org/wiki/List_of_regions_of_China

³https://en.wikipedia.org/wiki/Political_spectrum

Comment on news ($\phi_1()$)

News content n

User profile u : You are a Twitter user+You are {specific user attribute value}.

Prompt: u +You view a piece of news with the following content on Twitter.

News: $[n]$

Task: Please comment on this news. Your comment is limited to 40 words and must conform to your identity’s expression habits.

Your comment:

Reply to a comment ($\phi_2()$)

News content n

User profile u : You are a Twitter user+You are {specific user attribute value}.

Sampled node f

Comment chain C

Prompt: u +and a related comment chain on Twitter, and their contents are as follows:

News: $[n]$

Comment chain: $[f]$

Task: Please reply to the the last comment f on Twitter. Your reply is limited to 40 words and must conform to your identity’s expression habits.

Your reply:

B Experimental Setups

B.1 Details of Datasets

We conduct experiments on three fake news datasets, i.e., Twitter Lin *et al.* [2022], CED Song *et al.* [2019] and PHEME Zubiaga *et al.* [2016]. The statistics of the three datasets are shown in Table 2.

- **Twitter** contains tweets published on Twitter⁴, and each tweet is annotated with true or false.
- **CED** contains Chinese rumor data scraped from Weibo, including forwarding and comment information related to the original Weibo posts.

⁴In July 2023, Twitter has been rebranded to X.

User Attribute	Values
Age	under 18; 18–29; 30–49; 50–64; over 65
Gender	male; female
Education	college graduate; non-college graduate; high school or below
Politics	Republican; Democrat (<i>English</i>) Radical; neutral; conservative (<i>Chinese</i>)
Registration	<1 month; 1–6 months; 6 months–3 years; ≥ 3 years
Occupation	unemployed/student; education; administration; engineer; labor; logistics; medical; financial; media; arts
Location	Northeast; Midwest; South; West (<i>English</i>) North China; East China; Southwestern China; Northwestern China; South Central China; Northeast China (<i>Chinese</i>)
Verification	verified; non-verified

Table 1: Attribution assignments for synthetic propagation generation.

Algorithm 1 Synthetic Propagation Generation

Require: News content n ; graph size m ; probability α ; parameter β

Ensure: List of nodes V , edges E , mapping idx2node , user profiles

- 1: **Function** GENERATEUSERNEWSNETWORK
- 2: Initialize: $V = [0]$, $E = []$, $F = [n]$, $P = []$
- 3: Initialize: height/width of root = 1, $\text{idx2node}[0] = n$
- 4: **function** ψ
- 5: Generate a random user profile with attributes
- 6: **return** user profile and attributes
- 7: **end function**
- 8: **function** $\phi_1(n, u)$
- 9: Create comment prompt for user u on news n
- 10: **return** prompt
- 11: **end function**
- 12: **function** $\phi_2(n, u, f)$
- 13: Get comment chain C to f , create reply prompt
- 14: **return** prompt
- 15: **end function**
- 16: **while** $|V| \leq \text{graph size do}$
- 17: Generate user profile $u = \psi$
- 18: Draw random number $p \in [0, 1]$
- 19: **if** $p \leq \alpha$ **or** $|V| = 1$ **then**
- 20: Create comment prompt $\text{PROMPT} = \phi_1(n, u)$
- 21: Set $f = 0$
- 22: **else**
- 23: Filter non-root nodes $V_1 = V \setminus \{0\}$
- 24: Compute probabilities P based on height/width
- 25: Sample node f from V_1 using P
- 26: Create reply prompt $\text{PROMPT} = \phi_2(n, u, f)$
- 27: **end if**
- 28: Generate comment $s_{\text{out}} = \text{LTM}(\text{PROMPT})$
- 29: Add s_{out} to idx2node mapping
- 30: Append new node to V , edge $(f, \text{new node})$ to E
- 31: Update user profile list and node attributes
- 32: Update height and width of the new node
- 33: **end while**
- 34: **return** V , E , idx2node , user profiles

rumors released on Twitter during 5 emergency events between 2014 and 2016.

B.2 Details of Comparison Baselines

We compare InfoPDF with a total of 21 representative baselines, including 10 content-based methods and 11 propagation-based methods.

Content-based Methods. We include BERT Devlin *et al.* [2018] and DeBERTa He *et al.* [2020], two widely used pre-trained language models that encode news text and feed the final-layer representations into a classifier for fake news detection. EANN Wang *et al.* [2018] aims to learn event-invariant representations to improve generalization across events, and we adopt its text-only variant for a fair comparison. DEFEND Shu *et al.* [2019a] introduces a sentence-comment co-attention mechanism to model the interaction between news content and user comments. DualEmo Zhang *et al.* [2021] exploits both publisher emotion and social emotion, as well as their discrepancy, to enhance fake news detection. ARG Hu *et al.* [2024] incorporates LLM-generated reasoning signals to assist smaller models in making more accurate predictions, and we use its offline distillation version (ARG-D) in our experiments. GeFEND Nan *et al.* [2024] is a generative feedback-enhanced framework that leverages LLMs to synthesize comments from different user groups and aggregates them to improve detection performance. TELLER Liu *et al.* [2024] combines human-expertise-guided logic predicate generation with LLM-produced logic atoms and performs rule-based aggregation for explainable and generalizable detection. CAMERED Wang *et al.* [2025] integrates a mixture-of-experts architecture into a generative model to synthesize informative comments by simulating expert collaboration and controversy, alleviating semantic sparsity in the early stages of propagation. Finally, LLM_{text} and LLM_{comment} Chen *et al.* [2025] directly employ an LLM as the detector by prompting with the news content only or with both news and comments, respectively.

Propagation-based Methods. GCN Kipf [2016] models the news propagation graph using graph convolution operations to learn propagation-aware representations. GAT Veličković *et al.* [2018] applies graph attention mechanisms to capture informative neighbors and encode the propagation structure. GraphSAGE Hamilton *et al.* [2017] learns graph representations by sampling and aggregating features from local neighborhoods, enabling inductive inference. BiGCN Bian *et al.* [2020] constructs bidirectional propagation graphs and captures both top-down and bottom-up diffusion patterns for rumor detection. EBGCN Wei *et al.* [2021] models uncertain propagation structures using Bayesian graph convolution to improve robustness. UPSR Wei *et al.* [2022] reconstructs propagation representations with Gaussian modeling to capture potential structural information under uncertainty. UPFD Dou *et al.* [2021] jointly models news content and propagation structure, incorporating user preference features for improved detection. RAGCL Cui and Jia [2024] learns robust representations via adaptive propagation graph contrastive learning. DELL Wan *et al.* [2024] leverages LLMs to simulate diverse demographic attributes, generate

Datasets	Twitter	CED	PHEMES
Number of News	1,154	3,387	5,801
Number of True News	575	1,538	3,829
Number of False News	579	1,849	1,973
Number of Comments	59,255	1,275,180	85,408
Average number of Comments	52	377	15

Table 2: The statistics of datasets.

contextual explanations, and predict news with expert proxy tasks, where the final decision is obtained by aggregating multiple predictions. EIN Jiang *et al.* [2025] uses an LLM to annotate stance information for messages in the propagation tree and further models the propagation process from an epidemiology perspective. LLM_{propagation} Chen *et al.* [2025] directly prompts an LLM with propagation-related information (e.g., structured propagation context) to perform fake news detection.

B.3 Implementation Details

We utilize *Mistral-7B-Instruct-v0.2* Jiang *et al.* [2023]⁵ for Twitter and *Qwen1.5-7B-Chat* Yang *et al.* [2024]⁶ for CED in propagation generation. We set the temperature to 0.6 for *Mistral-7B-Instruct-v0.2* and use the default generation settings for *Qwen1.5-7B-Chat*. As for the text embedding, we employ a pre-trained BERT to encode the textual features of each news article and comment⁷. For the training of the encoders of different attributes, the dimension of the hidden vectors is set to 128, the training epochs are set to 100 and the learning rate is set to $2e-4$. We employ grid search to select the value of λ and β from $\{0.001, 0.01, 0.1, 1, 2\}$. We use PyTorch version 1.12.1 and Geometric version 2.3.1.

For LLM-based backbones, we employ *Mistral-7B-Instruct-v0.2* to detect fake news on Twitter and *Qwen1.5-7B-Chat* on CED. And for the proxy tasks in DELL, we removed the Knowledge Retrieval task. We implement other comparison methods under the same environment according to the parameter settings reported in their original papers. We select InfoPDF w/GraphSAGE for subsequent sub-experiments. We run five times and report the average results.

For baselines including LLM_{text}, LLM_{comments} and LLM_{propagation}, we implement them by the following prompts:

News-only Fake News Detection

Now you are an annotator to detect misinformation. You need to determine whether the given claim is misinformation. The claim will be given in the format [Claim [id, claim]]. Given the claim [Claim [id, claim]], please determine whether the claim is misinformation based on its comments and propagation structure. If the claim is true, you are supposed to answer 1; if it is false, you are supposed to answer 0. You have to choose one of the two answers, 0 or 1. No need to answer any other words except 0 or 1. Your answer is:

Comments-based Fake News Detection

Now you are an annotator to detect misinformation. You need to determine whether the given claim with comments is misinformation. The claim and its comments will be given in the format [Claim[id:claim]], [Comments [comment:texts]]. Given the claim [Claim [<id, claim>]], [Comments [<id, comments>]], please determine whether the claim is misinformation based on its comments. If the claim is true, you are supposed to answer 1; if it is false, you are supposed to answer 0. You have to choose one of the two answers, 0 or 1. No need to answer any other words except 0 or 1. Your answer is:

Propagation Structure-based Fake News Detection

Now you are an annotator to detect misinformation. You need to determine whether the given claim with comments and propagation structure composed of comments is misinformation. The claim, its comments, and the propagation structure composed of comments will be given in the format [Claim[id:claim]], [Comments [id:comment:texts]], [propagation structure: ['id1 replied to id2', ...]]. [id:comment:text] means the ID and text of the comments on the tweet, and 'id1 replied to id2' means the comment with ID id1 is a reply to the comment with ID id2.

Given the claim [Claim [<id, claim>]], [Comments [<id, comments>]], and [propagation structure [<propagation structure>]], please determine whether the claim is misinformation based on its comments and propagation structure. If the claim is true, you are supposed to answer 1; if it is false, you are supposed to answer 0. You have to choose one of the two answers, 0 or 1. No need to answer any other words except 0 or 1. Your answer is:

C Supplementary Experimental Results

We provide supplementary analysis of effects of different synthetic propagation size, semantic analysis of synthetic propagation, case study, and representation visualization.

C.1 Effect of Different Synthetic Propagation Generation Sizes

Figure 1 shows results against different sizes of LLM-generated synthetic propagation. InfoPDF consistently outperforms the baseline across all sizes, demonstrating the positive effect of incorporating synthetic propagation via our

⁵<https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.2>

⁶<https://huggingface.co/Qwen/Qwen1.5-7B-Chat>

⁷We employ *bert-base-uncased* for Twitter and *bert-base-chinese* for CED.

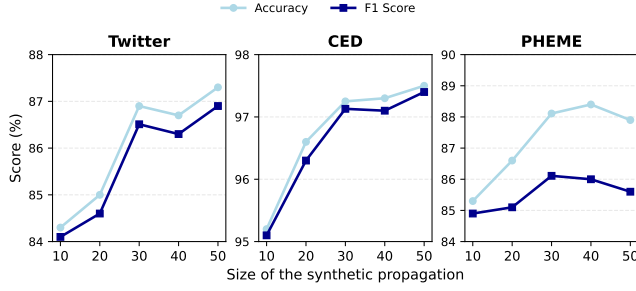


Figure 1: Results (%) against different synthetic propagation sizes on three datasets.

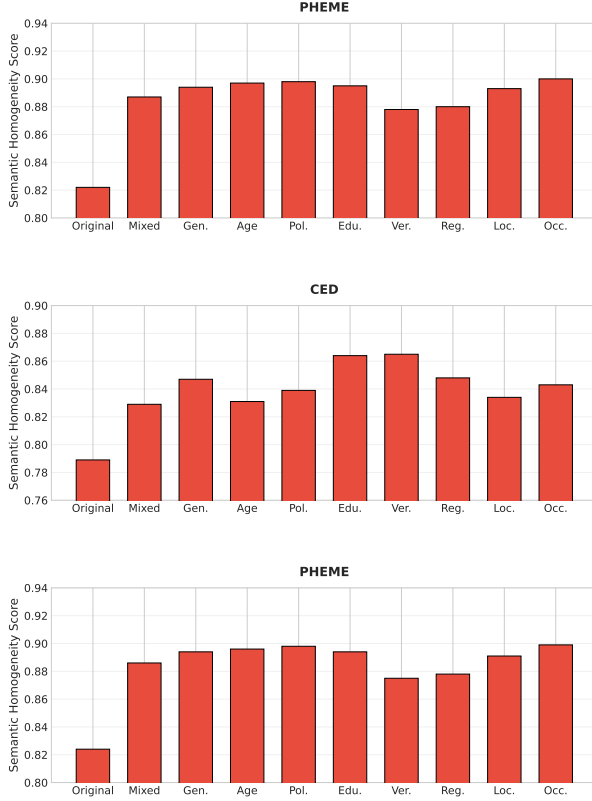


Figure 2: Diversity of different propagation.

framework.

Moreover, as the number of generated nodes increases, InfoPDF exhibits increasingly larger improvements on Twitter and CED. On PHEME, expanding the synthetic propagation provides limited performance gains when the size exceeds 30 nodes. This may be attributed to high topical diversity of the dataset.

C.2 Semantic Diversity Analysis of LLM-generated Synthetic Propagation

To evaluate the diversity of synthetic propagation, including original propagation in the dataset and synthetic propagation using different strategies, we calculate the semantic homogeneity score of the propagation. Specifically, we calculate

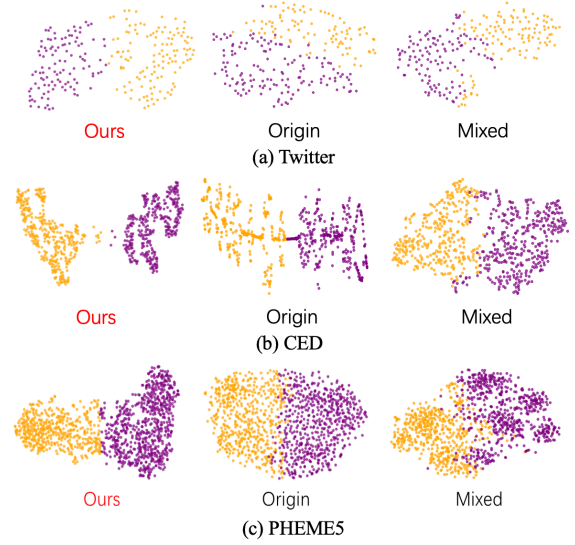


Figure 3: t-SNE visualization of representations learned based on different propagation data on the test set. Dots with the same color are within the same category. If the boundaries between different categories are clear and the distributions of samples within each class are tightly clustered, it indicates that the representations are more distinctive.

the mean of cosine similarity between comments based on their initial textual feature.

The results of diversity are shown in Figure 2. The similarity results for the original propagation are the lowest, indicating that the repetitiveness of its propagated content is lower. The relatively high homogeneity of the content in synthetic propagation indicates that its content is overly one-sided, leading to distorted propagation.

C.3 Representation Visualization

To intuitively demonstrate the impact of different propagation and the effect of InfoPDF, we feed the encoder (GCN) with different propagation and InfoPDF with the original propagation and synthetic propagation. we qualitatively visualize the learned representations of the test sets from datasets using t-SNE Van der Maaten and Hinton [2008].

Visualization results are shown in Figure 3. Compared with representation visualizations of real propagation and synthetic propagation generated with attribute-mixed strategy, our method exhibits better representation performance. The clusters of our method appear to be well-defined and distinct, which suggests that our method provides a richer feature set, allowing the encoder to capture more nuanced relationships and structures in the data.

C.4 Case Study

As shown in Figure 4, although there are some voices of skepticism about the news in the original propagation, they were not enough to make a significant impact. In Politics-specific synthetic propagation, there is a stronger skepticism and difference of opinion about the news. Meanwhile, the

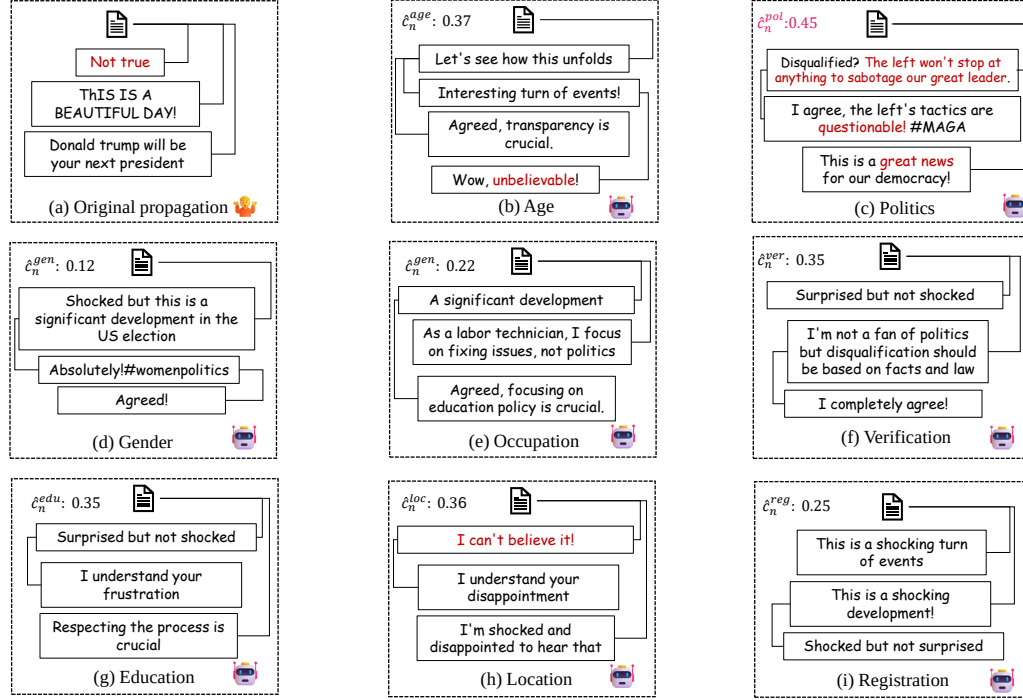


Figure 4: Specific content of each synthetic propagation. we select the most representative comments from each synthetic propagation for brevity.

synthetic propagation from the Age and Location perspectives also contain some skepticism towards the news.

References

- Tian Bian, Xi Xiao, Tingyang Xu, et al. Rumor detection on social media with bi-directional graph convolutional networks. In *AAAI*, volume 34, pages 549–556, 2020.
- Mengyang Chen, Lingwei Wei, Han Cao, Wei Zhou, and Songlin Hu. Explore the potential of llms in misinformation detection: An empirical study. In *AAAI 2025 Workshop on Preventing and Detecting LLM Misinformation (PDLIM)*, 2025.
- Chaoqun Cui and Caiyan Jia. Propagation tree is not deep: Adaptive graph contrastive learning approach for rumor detection. In *Proceedings of the AAAI Conference on artificial intelligence*, volume 38, pages 73–81, 2024.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018.
- Yingtong Dou, Kai Shu, Congying Xia, et al. User preference-aware fake news detection. In *SIGIR*, pages 2051–2055, 2021.
- Will Hamilton, Zhitao Ying, and Jure Leskovec. Inductive representation learning on large graphs. *NIPS*, 30, 2017.
- Pengcheng He, Xiaodong Liu, Jianfeng Gao, and Weizhu Chen. Deberta: Decoding-enhanced bert with disentangled attention. *ArXiv*, abs/2006.03654, 2020.

- Beizhe Hu, Qiang Sheng, Juan Cao, Yuhui Shi, Yang Li, Danding Wang, and Peng Qi. Bad actor, good advisor: Exploring the role of large language models in fake news detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 22105–22113, 2024.
- Albert Q Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. Mistral 7b. *arXiv preprint arXiv:2310.06825*, 2023.
- Wei Jiang, Tong Chen, Xinyi Gao, Wentao Zhang, Lizhen Cui, and Hongzhi Yin. Epidemiology-informed network for robust rumor detection. In *Proceedings of the ACM on Web Conference 2025*, pages 3618–3627, 2025.
- TN Kipf. Semi-supervised classification with graph convolutional networks. *arXiv preprint arXiv:1609.02907*, 2016.
- H. Lin, J. Ma, L. Chen, Z. Yang, M. Cheng, and C. Guang. Detect rumors in microblog posts for low-resource domains via adversarial contrastive learning. In *Proc. NAACL-HLT*, pages 2543–2556, 2022.
- Hui Liu, Wenya Wang, Haoru Li, and Haoliang Li. TELLER: A trustworthy framework for explainable, generalizable and controllable fake news detection. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Findings of the Association for Computational Linguistics: ACL 2024*, pages 15556–15583, Bangkok, Thailand, August 2024. Association for Computational Linguistics.
- Abdul Motaleb, Aakib Bin Nesar, Joytun Nur Sultana,

245 Md Aminul Islam, Iftekhar Mahmud Tahir, Md Mansur
246 Ahmed, and Rashedur M Rahman. Analyzing human abili-
247 ties to detect fake bangla news with respect to different fea-
248 tures. In *2021 IEEE 12th Annual Ubiquitous Computing,
249 Electronics & Mobile Communication Conference (UEM-
250 CON)*, pages 0337–0346. IEEE, 2021.

251 Qiong Nan, Qiang Sheng, Juan Cao, Beizhe Hu, Danding
252 Wang, and Jintao Li. Let silence speak: Enhancing fake
253 news detection with generated comments from large lan-
254 guage models. In *Proceedings of the 33rd ACM Inter-
255 national Conference on Information and Knowledge Man-
256 agement*, pages 1732–1742, 2024.

257 Kai Shu, Limeng Cui, Suhang Wang, Dongwon Lee, and
258 Huan Liu. defend: Explainable fake news detection. In
259 *Proceedings of the 25th ACM SIGKDD international con-
260 ference on knowledge discovery & data mining*, pages 395–
261 405, 2019.

262 Kai Shu, Xinyi Zhou, Suhang Wang, Reza Zafarani, and
263 Huan Liu. The role of user profiles for fake news detec-
264 tion. In *Proceedings of the 2019 IEEE/ACM international
265 conference on advances in social networks analysis and
266 mining*, pages 436–439, 2019.

267 Changhe Song, Cheng Yang, Huimin Chen, et al. Ced:
268 Credible early detection of social media rumors. *TKDE*,
269 33(8):3035–3047, 2019.

270 Laurens Van der Maaten and Geoffrey Hinton. Visualizing
271 data using t-sne. *Journal of machine learning research*,
272 9(11), 2008.

273 Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adri-
274 ana Romero, Pietro Liò, and Yoshua Bengio. Graph atten-
275 tion networks. In *International Conference on Learning
276 Representations*, 2018.

277 Herun Wan, Shangbin Feng, Zhaoxuan Tan, Heng Wang, Yu-
278 lia Tsvetkov, and Minnan Luo. Dell: Generating reactions
279 and explanations for llm-based misinformation detection.
280 *arXiv preprint arXiv:2402.10426*, 2024.

281 Yaqing Wang, Fenglong Ma, Zhiwei Jin, Ye Yuan, Guangxu
282 Xun, Kishlay Jha, Lu Su, and Jing Gao. Eann: Event ad-
283 versarial neural networks for multi-modal fake news detec-
284 tion. In *Proceedings of the 24th acm sigkdd international
285 conference on knowledge discovery & data mining*, pages
286 849–857, 2018.

287 Bing Wang, Bingrui Zhao, Ximing Li, Changchun Li, Wanfu
288 Gao, and Shengsheng Wang. Collaboration and contro-
289 versy among experts: Rumor early detection by tuning a
290 comment generator. In *Proceedings of the 48th Interna-
291 tional ACM SIGIR Conference on Research and Develop-
292 ment in Information Retrieval*, pages 468–478, 2025.

293 Lingwei Wei, Dou Hu, Wei Zhou, et al. Towards propagation
294 uncertainty: Edge-enhanced Bayesian graph convolutional
295 networks for rumor detection. In *ACL*, pages 3845–3854,
296 August 2021.

297 Lingwei Wei, Dou Hu, Wei Zhou, et al. Uncertainty-aware
298 propagation structure reconstruction for fake news detec-
299 tion. In *COLING*, pages 2759–2768, 2022.

An Yang, Baosong Yang, Binyuan Hui, Bo Zheng, Bowen Yu,
Chang Zhou, Chengpeng Li, Chengyuan Li, Dayiheng Liu,
Fei Huang, et al. Qwen2 technical report. *arXiv preprint
arXiv:2407.10671*, 2024.

Xueyao Zhang, Juan Cao, Xirong Li, Qiang Sheng, Lei
Zhong, and Kai Shu. Mining dual emotion for fake news
detection. In *Proceedings of the web conference 2021*,
pages 3465–3476, 2021.

Arkaitz Zubiaga, Maria Liakata, and Rob Procter. Learning
reporting dynamics during breaking news for rumour de-
tection in social media. *arXiv preprint arXiv:1610.07363*,
2016.